



Menopausal hormone therapy: Benefits and risks

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INTRODUCTION

The mean age of menopause is 51 years, with 95 percent becoming menopausal between the ages of 45 to 55 years. During the menopause transition and postmenopause, most women experience vasomotor symptoms (hot flashes and night sweats). Systemic estrogen is the most effective treatment available for relief of hot flashes. Other symptoms associated with perimenopause and menopause that respond to systemic estrogen include sleep disturbances, mood lability/depression, and, in some cases, joint aches and pains. Postmenopausal women may also experience vulvovaginal symptoms including dyspareunia (genitourinary syndrome of menopause) which can be effectively treated by local administration of estrogen.

An overview of the risks and benefits of menopausal hormone therapy (MHT) will be provided here. The clinical features and diagnosis of menopause, as well as the management of women with hot flashes and other menopausal symptoms are reviewed in greater detail separately. The management of women with early menopause and primary ovarian insufficiency are also reviewed separately.

- (See "[Menopause: Clinical features and diagnosis](#)".)
- (See "[Menopausal hot flashes](#)".)
- (See "[Treatment of menopausal symptoms with hormone therapy](#)".)
- (See "[Genitourinary syndrome of menopause: Treatment of vulvovaginal atrophy](#)".)
- (See "[Management of primary ovarian insufficiency \(premature ovarian failure\)](#)".)

ESTIMATES OF OVERALL RISK BASED ON AGE

Individualized risk assessment — We agree with most professional societies that advise not starting MHT without an individualized assessment of the risks and benefits [1-4]. When counseling symptomatic women considering MHT, the clinician should provide the best estimate of potential risks and benefits for up to five years of use based upon age-specific data [2,5,6]. (See "[Treatment of menopausal symptoms with hormone therapy](#)", section on '[Calculating risks](#)'.)

Vasomotor symptoms (hot flashes) are the most common indication for the use of MHT. As a result, almost all women who seek medical therapy for menopausal symptoms do so in their late 40s or 50s. MHT is no longer recommended for prevention of chronic disease such as coronary heart disease (CHD) or osteoporosis, preservation of cognitive function, or prevention of dementia. (See "[Treatment of menopausal symptoms with hormone therapy](#)", section on '[Clinical indications](#)'.)

Women ≥ 60 years — We agree with most professional societies that advise individualizing the use of MHT, particularly in women 60 to 69 years of age, since the benefit-risk ratio is less favorable in women ≥ 60 years than in women 50 to 59 years [1-3]. We avoid MHT in women ≥ 70 years [7]. (See '[Coronary heart disease](#)' below and "[Treatment of menopausal symptoms with hormone therapy](#)", section on '[Calculating risks](#)' and "[Treatment of menopausal symptoms with hormone therapy](#)", section on '[Extended use of MHT](#)'.)

The American Geriatric Society advises not starting MHT in women ≥ 65 years [4], since in the Women's Health Initiative (WHI) and subsequent meta-analyses, there was an excess risk of vascular events with [conjugated equine estrogen](#) (with or without [medroxyprogesterone acetate](#)) compared with placebo [5,8-11]. The WHI reported adverse effects of menopausal MHT in older postmenopausal women (≥ 60 years or more than 10 years since menopause); this is not the age group that presents with new onset of menopausal symptoms. In addition, the data in the WHI are based on a hormone regimen that is no longer a preferred regimen. (See '[Cardiovascular effects](#)' below and "[Treatment of menopausal symptoms with hormone therapy](#)", section on '[Our preferred regimens](#)'.)

Women 50 to 59 years — Most expert societies suggest MHT for younger postmenopausal women with moderate to severe symptoms and no contraindications to estrogen use [1,2,12]. Women should be reassured that the absolute risk of complications for healthy, young postmenopausal women taking MHT is very low ([figure 1](#)). (See "[Treatment of menopausal symptoms with](#)



Although most available MHT clinical trial data are from women over age 60 years, estimates of the absolute risks and benefits for women starting MHT in their 50s has been published ([figure 1](#)) [2,5,6]. The intervention phase of the WHI was the source of data for the analysis, and these data are based on one MHT regimen ([conjugated equine estrogen](#) [CEE] and [medroxyprogesterone acetate](#)). Alternative MHT regimens (eg, transdermal [estradiol](#) and micronized [progesterone](#)) may have a superior safety profile. (See '[Cardiovascular effects](#)' below.)

Figure 1 - larger image below

The data below are expressed as the attributable excess risk or benefit for five years of combined estrogen-progestin or unopposed estrogen use in women starting treatment between ages 50 to 59 years. Overall, the risk-benefit profile was more favorable for women ages 50 to 59 compared with older women [2,5].

- **Combined estrogen-progestin therapy** – Number of cases (additional or fewer) per 1000 women per five years of hormone use when compared with placebo ([figure 1](#)):
 - Coronary heart disease (CHD) – 2.5 additional cases
 - Invasive breast cancer – 3 additional cases
 - Stroke – 2.5 additional cases
 - Pulmonary embolism – 3 additional cases
 - Colorectal cancer – 0.5 fewer cases
 - Endometrial cancer – No difference
 - Hip fracture – 1.5 fewer cases
 - All-cause mortality – 5 fewer events
- **Estrogen-alone therapy** – Number of cases (additional or fewer) per 1000 women per five years of hormone use when compared with placebo ([figure 1](#)):
 - CHD – 5.5 fewer cases
 - Invasive breast cancer – 2.5 fewer cases
 - Stroke – 0.5 fewer cases
 - Pulmonary embolism – 1.5 additional cases
 - Colorectal cancer – 0.5 fewer cases
 - Hip fracture – 1.5 additional cases (of note, there was an overall decrease in all osteoporotic fractures in both the estrogen and combined estrogen-progestin groups)
 - All-cause mortality – 5.5 fewer events

This analysis highlights that the overall risks of MHT in younger postmenopausal women are considerably lower than those for older women (eg, women in the WHI). The major

explanation for the difference in absolute excess risk between older and younger postmenopausal women is the lower baseline risk of CHD, stroke, venous thromboembolism (VTE), and breast cancer in younger postmenopausal women [5,6]. (See '[Cardiovascular effects](#)' below and '[Breast cancer](#)' below.)

A comprehensive overview of findings of the intervention and postintervention phases of the WHI (a total of 13 years of follow-up) highlighted differences in the risk-benefit profiles between combined CEE-medroxyprogesterone acetate and unopposed CEE therapy, as well as the impact of age on the effects of HT [5]. As summarized throughout this topic, combined therapy using CEE with [medroxyprogesterone acetate](#) was associated with a higher risk of CHD and breast cancer than unopposed estrogen.

CARDIOVASCULAR EFFECTS

Coronary heart disease — MHT is not recommended for primary or secondary prevention of coronary heart disease [8,9,13-15].

- **Older menopausal women** – In the Women's Health Initiative (WHI), the overall rate of coronary events was increased with combined CEE-medroxyprogesterone acetate therapy 2.3 versus 1.9 percent with placebo (hazard ratio [HR] 1.23, 95% CI 0.99-1.53). The estimated absolute excess risk was eight additional events per 10,000 person-years (95% CI 0.3 to 16) [16,17]. In a secondary analysis evaluating MHT in women with vasomotor symptoms by age group, women ≥70 years had increased risk of coronary events (HR 3.22, 95% CI 1.36-7.63; 382 excess events per 10,000 person years) [7]. However, in women aged 60 to 69 years, CEE-medroxyprogesterone acetate did not increase risk (HR 0.84, 95% CI 0.51-1.39; 40 fewer events per 10,000 person-years).

In the unopposed estrogen trial, use of CEE did not appear to affect the incidence of CHD events over an average follow-up of 6.8 years (3.8 versus 4 percent with placebo, HR 0.95, 95% CI 0.78-1.16 [estimated absolute risk reduction of three fewer events per 10,000 person-years]) [9,16]. In the secondary analysis by age group, the risk of coronary events in women with vasomotor symptoms who were ≥70 years was increased with CEE (HR 1.95, 95% CI 1.06-3.59; 217 excess events per 10,000 person-years) [7]. In women 60 to 69 years, the risk was increased but did not reach statistical significance (annualized percent 3 versus 2.3 percent with placebo, HR 1.31, 95% CI 0.9-1.9; 74 excess events per 10,000 person-years).

These data support an individualized risk assessment, particularly in women 60 to 69 years of age. (See '[Women ≥60 years](#)' above.)

- **Younger menopausal women** – Data from a coronary angiographic study [18], a WHI coronary artery calcification study and age-based subgroup analysis [16,19], and a 2006

meta-analysis that included the WHI [20] all suggest that the timing of treatment initiation affects the risk of CHD; in the WHI, women who were <10 years since menopause or between the ages of 50 to 59 years did not have excess risk [7,16] or possibly had a reduction in risk [20].

Additional reassuring data come from clinical trials in younger postmenopausal women:

- The Kronos Early Estrogen Prevention Study (KEEPS), a four-year, randomized, double-blind, placebo-controlled trial (in women ages 45 to 54 years), reported that, when combined with cyclical monthly oral [progesterone](#), oral CEE (0.45 mg daily) or transdermal estrogen (50 mcg daily) reduced menopausal symptoms. Surrogate markers of atherosclerosis (coronary artery calcium and carotid intima-media thickness [CIMT]) were not significantly different in the hormone therapy (HT) and placebo groups [21].
- In the Early versus Late Intervention Trial with [Estradiol](#) (ELITE), 643 postmenopausal women, stratified according to time since menopause (<6 or >10 years; early versus late, respectively), received oral estradiol (with [progesterone](#) for women with a uterus) or placebo for a median of five years [22]. Progression of subclinical atherosclerosis (measured as CIMT) was slower than placebo in the early intervention group, while rates of progression were similar to placebo in the late intervention group. Estradiol had no effect on computed tomography (CT) coronary artery calcium in either the early or late intervention group.
- A 2015 meta-analysis of 19 trials of oral (including the WHI), but not transdermal, MHT in over 40,000 postmenopausal women performed subgroup analyses in women who started MHT less than 10 years after menopause [23]. Results included the following:
 - A lower risk of CHD (composite of death from cardiovascular causes and nonfatal myocardial infarction) compared with placebo (relative risk [RR] 0.52, 95% CI 0.29-0.96; eight fewer cases of heart disease per 1000 women treated per year).
 - A lower mortality rate (RR 0.70, 95% CI 0.52-0.95; six fewer deaths per 1000 women treated per year).
 - The quality of the evidence for the mortality and CHD meta-analyses was rated as moderate (rated down one level because DOPS, a methodologically flawed trial [24], was included). When DOPS was removed, the analyses were no longer significant.

Stroke

- **Oral estrogen** – Clinical trial data suggest that oral estrogen increases stroke risk [25-28]. However, in women ages 50 to 59 years, the absolute risk is extremely low [16].

As examples:

- In the WHI, a 31 percent increase in stroke risk was seen with combined CEE-medroxyprogesterone acetate use compared with placebo (intention-to-treat HR 1.31, 95% CI 1.02-1.68) ([figure 2](#)). The increase in risk is for ischemic but not hemorrhagic stroke [25]. Excess risk was seen in all age groups [16] and was independent of other known risk factors for stroke. In the unopposed estrogen trial, stroke risk was significantly increased with CEE versus placebo (HR 1.39, 95% CI 1.1-1.77) [9]. In a subgroup analysis that combined data from both trials, the RR of stroke did not vary by patient age. Because the baseline risks are lower in younger postmenopausal women, the calculated absolute risk for women ages 50 to 59 years age is low (0.15 versus 0.13 cases per 100 women per year for MHT and placebo, respectively) [16].
- In meta-analyses of randomized trials (including WHI), oral estrogen therapy (with or without a progestogen) was associated with an increased risk of stroke overall (eg, 2.6 versus 2.1 percent with placebo or no treatment, risk ratio 1.23, 95% CI 1.08-1.41) [10,29]. In subgroup analysis (MHT within 10 years of menopause versus >10 years after menopause), there were no subgroup differences for stroke [29]. However, the absolute risk of stroke was lower in women <10 years from menopause (1.1 versus 4 percent for women ≥10 years since menopause) [29].
- **Transdermal estrogen** – Low-dose, transdermal estrogen therapy does not appear to increase the risk of stroke in postmenopausal women. This was illustrated in a population-based, nested case-control study that included approximately 16,000 cases of stroke and 60,000 controls [28]. The overall rate of stroke in the cohort was 2.85 per 1000 per year. The risk of stroke was not increased in low-dose (≤50 mcg patch) transdermal estrogen users compared with nonusers (rate ratio 0.81, 95% CI 0.62-1.05) but was increased with higher transdermal doses (>50 mcg patch; 1.89, 95% CI 1.15-3.11).

Venous thromboembolism

- **Oral estrogen** – Oral estrogen increases the risk of venous thromboembolism (VTE) [10,13,29,30]. However, in women ages 50 to 59 years, the absolute risk is lower than in older postmenopausal women.

In the WHI, the estimates of excess VTE risk overall (all age groups) with was 9 and 4 per 1000 over five years for combined [conjugated equine estrogen](#) (CEE)-medroxyprogesterone acetate therapy and CEE alone, respectively [6,30]. For women ages 50 to 59 years, the group most likely to be taking MHT, the estimates of excess VTE risk were approximately 5 and 2 additional cases per 1000 women per five years of combined estrogen-progestin or CEE alone, respectively [6]. (See '[Women 50 to 59 years](#)' above.)

- **Transdermal estrogen** – The risk of VTE appears to be lower with transdermal compared with oral estrogen preparations [31,32]. In a meta-analysis of observational studies, no

excess risk of VTE was observed in women taking transdermal estrogen (OR 1.2, 95% CI 0.1-1.7), even in those with prothrombotic mutations or high body mass index (BMI) [31]. Similar results were reported in a population-based cohort study published after the meta-analysis [33].

- **Type of progestogen** – In addition to the route of estrogen administration, risk may vary by type of progestogen (higher with [medroxyprogesterone acetate](#), likely lower with micronized [progesterone](#)) [34]. In a study of over one million postmenopausal women that included 2200 venous thromboembolic events, the relative risk of VTE in current hormone users versus nonusers was higher for women taking oral estrogen-progestogen than unopposed estrogen regimens (RR 2.07 versus 1.42) [34]. Transdermal estrogen users had no excess risk. Among the oral estrogen-progestogen users, risk was greater for regimens containing medroxyprogesterone acetate than other progestogens (RR 2.67 versus 1.91).

Lipids — Oral estrogen has a proven beneficial effect on serum lipid concentrations [35-37], which can be negated in part by some progestogens such as [medroxyprogesterone acetate](#). Micronized [progesterone](#) has a neutral effect. These effects appear to be independent of age, with similar results noted in younger and older postmenopausal women [37].

The effect of oral [estradiol](#) (1 mg/day) is similar to that of oral CEE 0.625 mg, while transdermal estradiol has a lesser effect than oral estrogen preparations [35]. The lack of effect with transdermal estrogen presumably reflects decreased exposure of the liver to estrogen (compared with oral therapy) due to avoidance of the first-pass effect on the liver. However, another study which treated women for much longer with varying doses of transdermal estradiol found that the serum total and LDL cholesterol concentrations decreased in a dose-dependent fashion after two years of therapy [38].

Blood pressure — Replacement doses of estrogen have little effect on blood pressure. The Women's Health Initiative (WHI) combined estrogen-progestin trial noted a small increase (1.5 mmHg) in systolic pressure compared with placebo [8]; a similar difference between the hormone and placebo groups of 1.1 mmHg was noted in the WHI trial of unopposed estrogen [9].

MORTALITY

Available data suggest that MHT is not associated with an increase in mortality rates when compared with placebo. Younger menopausal women (ages 50 to 59) may actually have a reduction in all-cause mortality with MHT [39]. Early epidemiologic studies reported a reduction in mortality rates with MHT [8,9,16,39,40]. After excess cardiovascular risks were reported in early Women's Health Initiative (WHI) reports, there were concerns that MHT

could actually be associated with an increase in cardiovascular mortality. However, this does not appear to be the case.

The best available data come from an 18-year follow-up report from the WHI trials of unopposed estrogen (median duration seven years) or estrogen-progestin therapy (mean duration five years) versus placebo [39]. All-cause mortality was similar (approximately 27 percent) in women taking MHT or placebo. Cardiovascular and cancer (including breast cancer) mortality rates were also similar for MHT and placebo. When analyzed by 10-year age groups, MHT reduced all-cause mortality in women ages 50 to 59 years during the intervention phase (hazard ratio [HR] 0.61, 95% CI 0.43-0.87) and possibly after 18 years of follow-up, although the reduction was no longer statistically significant (HR 0.87, 95% CI 0.76-1.0).

In an earlier analysis, women ages 50 to 59 years, the group most likely to be taking MHT, the estimate of mortality benefit in one analysis was 5.3 and 5 fewer deaths per 1000 per five years of combined estrogen-progestin or unopposed estrogen use, respectively [6]. (See ['Women 50 to 59 years'](#) above.)

Other evidence for the effect of MHT on mortality comes from a meta-analysis of 19 randomized controlled trials in younger postmenopausal women, including age-specific data (women 50 to 59 years) from the two WHI trials and 17 trials with a mean subject age <60 years [41]. In the pooled analysis of approximately 16,000 women (mean age 55 years) followed for an average of 5.1 years, MHT reduced mortality by 27 percent when compared with placebo (relative risk [RR] 0.73, 95% CI 0.52-0.96). The absolute reduction in mortality was 0.84 percent (ie, 1 in every 119 women treated with HT did not die at five years compared with untreated patients).

These data provide additional reassurance that HT is a safe option for the treatment of symptoms in younger postmenopausal women [42]. (See ['Women 50 to 59 years'](#) above and ["Treatment of menopausal symptoms with hormone therapy"](#).)

CANCER

Breast cancer — In the WHI overall study population, combined CEE-medroxyprogesterone acetate use compared with placebo increased the risk of breast cancer (absolute excess risk of 9 cases per 10,000 person years [HR 1.24, 95% CI 1.01-1.53]) [5]. However, the absolute risk of breast cancer with combined MHT is very low with typical duration of use in postmenopausal women ages 50 to 59 years (eg, three additional cases per 1000 women with CEE plus [medroxyprogesterone acetate](#) for five years, therefore <1 per 1000/year) ([figure 1](#)) [1,2,5]. The magnitude of this risk is similar to reported risks in women with obesity and low physical activity [1]. There does not appear to be an additive effect of

hormone therapy with age or underlying breast cancer risk (eg, genetic risk of breast cancer, family history, personal breast cancer risk factors) on breast cancer incidence. In addition, hormone therapy use does not further increase after bilateral salpingo-oophorectomy (BSO) for BRCA 1 or 2 genetic variants [1]. Although the relative risk of breast cancer with hormone therapy is similar for all women, the absolute risk is higher for women with higher baseline risk for breast cancer.

In the WHI overall study population, unopposed estrogen (CEE) reduced the risk of breast cancer [5,9]. For women in their 50s taking unopposed CEE, there were an estimated 2.5 fewer cases per 1000 women per five years of hormone use when compared with placebo ([figure 1](#)) [1,2]. A 2024 meta-analysis of randomized trials (over 14,000 participants) showed a decreased risk of breast cancer with use of estrogen alone (3.6 versus 4.7 percent for estrogen alone and placebo, respectively, RR 0.77, 95% CI 0.65-0.91) [43]. In observational studies with longer duration and a meta-analysis of observational studies, use of unopposed CEE was associated with a small but statistically significant increase in breast cancer risk (20-year risk of breast cancer [between the ages of 50 and 69 years] by approximately one in every 200 users of estrogen-only regimens) [44,45].

Current MHT regimens typically include lower doses of estrogen as well as a different type and route (eg, transdermal 17-beta [estradiol](#) [17-beta-E2]). In addition a synthetic progestin, [medroxyprogesterone acetate](#), was used in the WHI trial, whereas micronized [progesterone](#) is now considered the progestogen of choice by most experts [2,12,46,47]. Limited observational data suggest that natural micronized progesterone may not be associated with additional risk of breast cancer [48]. It is thought that transdermal 17-beta-E2 and micronized progesterone may be associated with lower breast cancer risk than CEE and medroxyprogesterone acetate, although data are not yet conclusive. (See "[Preparations for menopausal hormone therapy](#)", section on 'Progestin preparations'.)

We advise against MHT use in breast cancer survivors, given a potentially heightened risk of recurrence associated with MHT. (See "[Overview of long-term complications of therapy in breast cancer survivors and patterns of relapse](#)", section on 'Menopausal symptoms'.)

Ovarian cancer — We do not consider ovarian cancer to be a major consideration when deciding to take MHT for symptomatic relief, because the absolute risk of ovarian cancer with MHT is very low. A nonsignificant increase in the risk of ovarian cancer was observed in the WHI with combined estrogen-progestin therapy (HR 1.6, 95% CI 0.8-3.2; 42 versus 27 cases per 100,000 person-years in the hormone and placebo groups, respectively) [49]. The excess risk was not statistically significant and would be approximately 0.75 cases per 1000 women treated for five years [6]. There were no differences in the distributions of tumor grade, stage, or histology. Because of the small number of ovarian cancer cases and the limited precision in estimating effects in the trial, these results should not affect a woman's decision to take MHT for symptomatic relief.

A meta-analysis of 52 epidemiologic studies including 21,488 postmenopausal women with ovarian cancer suggests that there is a small excess risk of ovarian cancer with MHT use [50]. Ovarian cancer risk was greater in ever-users versus never-users of MHT (relative risk [RR] 1.14, 95% CI 1.10-1.19). Risks were similar for estrogen-only and estrogen-progestin users. The calculated absolute excess risk associated with MHT was very low; five years of MHT use in women ages 50 to 54 years would result in approximately one additional ovarian cancer case per 1000 users and one ovarian cancer death per 1700 users. Age at initiation had little effect on RR, nor did smoking, body mass index, or past use of oral contraceptives.

In one case-control study of 162 matched sets of women with *BRCA1* or *BRCA2* mutations, MHT did not appear to increase the risk of ovarian cancer; odds ratio (OR) with every use of HT was 0.93 (95% CI 0.56-1.56) [51].

Risk factors for ovarian cancer are discussed in detail elsewhere. (See "[Epithelial carcinoma of the ovary, fallopian tube, and peritoneum: Incidence and risk factors](#)".)

Endometrial hyperplasia and carcinoma — In patients with a uterus, systemic estrogen therapy from any route (oral, transdermal patch, vaginal ring) without concurrent administration of a progestogen increases the risk of developing endometrial hyperplasia and carcinoma. (See "[Endometrial carcinoma: Epidemiology, risk factors, and prevention](#)", section on 'Chronically increased estrogen levels or estrogenic activity' and "[Endometrial hyperplasia or endometrial intraepithelial neoplasia: Clinical features, diagnosis, and differential diagnosis](#)", section on 'Risk factors'.)

Equivalent doses of transdermal and oral estrogen appear to have similar effects on the endometrium. However, in one study, ultra low-dose transdermal estrogen (14 mcg) and placebo showed similar low rates of endometrial proliferation and endometrial hyperplasia [52]. Recommendations for progestogen administration vary. (See "[Treatment of menopausal symptoms with hormone therapy](#)", section on 'Progestogens'.)

Among women treated with estrogen, the excess risk of endometrial hyperplasia and carcinoma can be largely abolished by concurrent therapy with a progestogen given in either a cyclic or continuous regimen [49,53-56]. In the estrogen-progestin WHI trial, more women in the MHT group required endometrial biopsies to assess vaginal bleeding (33 versus 6 percent for placebo), but there was no significant difference in endometrial cancer risk between the two groups (HR 0.81, 95% CI 0.48-1.36) [49]. In the PEPI trial and in a review from the Cochrane Database, cyclic progestin (when given at least 12 days per month) was as effective as continuous low-dose progestin [55,57]. Shorter-duration progestin therapy (<10 days) may be less protective [57-62]. (See "[Treatment of menopausal symptoms with hormone therapy](#)", section on 'Progestogens'.)

Colorectal cancer — Initial data from WHI suggested a decrease in the risk of colorectal cancer with combined CEE-medroxyprogesterone acetate use but no effect with estrogen alone [63]. In a subsequent analysis with 23 years of follow-up, risks were similar in CEE + [medroxyprogesterone acetate](#), CEE alone, and placebo groups [64]. This topic is discussed in greater detail separately. (See "[Epidemiology and risk factors for colorectal cancer](#)", [section on 'Menopausal hormone therapy'](#).)

Lung cancer — Women on combined estrogen-progestin therapy who developed non-small cell lung cancer (NSCLC) had a significantly shorter survival compared with those given placebo. (See "[Females and lung cancer](#)", [section on 'Outcome'](#).)

COGNITIVE FUNCTION AND DEMENTIA

There is limited epidemiologic support for the hypothesis that estrogen preserves overall cognitive function in women. Meta-analyses of epidemiologic data [65] and randomized trials [66] have shown no benefit for prevention of dementia and no evidence of accelerated cognitive decline [67]. It is possible that there are domain-specific cognitive benefits (such as verbal memory) with MHT in younger postmenopausal women, but definitive evidence is lacking [68]. This topic is reviewed in more detail separately. (See "[Estrogen and cognitive function](#)".)

GALLBLADDER DISEASE

A secondary analysis of data from the Women's Health Initiative (WHI) found a significantly increased risk of biliary tract disease among women using oral estrogen therapy [69]. Both trials showed a greater risk of any gallbladder disease or surgery in the groups randomized to oral estrogen (annual incidence rate 78 versus 47 per 10,000 person-years for estrogen alone and placebo, respectively, and 55 and 35 per 10,000 person-years in the estrogen plus progestin and placebo groups, respectively). In a subsequent analysis of women ages 50 to 59 (or <10 years menopausal), the attributable excess risk of cholecystitis was calculated to be 9.6 and 14.2 additional cases per 1000 women per five years of combined estrogen-progestin therapy or unopposed estrogen use, respectively [6].

Similar results were noted in the Heart and Estrogen/Progestin Replacement Study (HERS) trial, in which 2763 postmenopausal women with coronary heart disease (CHD) were randomly assigned to combined estrogen-progestin therapy or placebo [70]. After an average follow-up of four years, there was a marginally significant increase in biliary tract surgery in the treatment group (38 percent). The authors calculated that for every 185 women receiving hormone therapy (HT), one additional woman had biliary tract surgery per year. (See "[Gallstones: Epidemiology, risk factors and prevention](#)", [section on 'Medications'](#).)

OSTEOPOROTIC FRACTURE

In the WHI trial, both unopposed estrogen and combined estrogen-progestin treatment reduced fracture risk. (See "[Menopausal hormone therapy in the prevention and treatment of osteoporosis](#)".)

For women ages 50 to 59 years, the group most likely to be taking MHT, the estimates of benefit in one analysis were 4.9 and 5.9 fewer fractures per 1000 women per five years of combined estrogen-progestin or unopposed estrogen use, respectively [6]. (See '[Women 50 to 59 years](#)' above.)

TYPE 2 DIABETES MELLITUS

Although MHT appears to reduce the risk of type 2 diabetes mellitus in women, the effect is insufficient to recommend MHT as a diabetes prevention strategy.

In a post hoc analysis from the Heart and Estrogen/Progestin Replacement Study (HERS), a secondary prevention of coronary heart disease (CHD) study, 734 of 2783 women had diabetes at baseline; the remaining 2029 women (who were normoglycemic or had impaired glucose tolerance) were followed for the development of type 2 diabetes for an average of four years [71]. Results of the trial were as follows:

- The cumulative incidence of type 2 diabetes was 6.2 percent in the hormone therapy (HT) group compared with 9.5 percent in the placebo group (relative hazard [RH] 0.65, 95% CI 0.48-0.89).
- Results were still significant after adjustment for variables such as body mass index, weight change, and waist circumference.
- The number needed to treat to prevent one case of diabetes was 30 (95% CI 18-103).

Similar results were reported in the Women's Health Initiative (WHI) combined estrogen-progestin trial in which fasting glucose and insulin were measured in a random sample at baseline and at one and three years. After a mean follow-up of 5.6 years, the following results were seen [72]:

- The cumulative incidence of treated diabetes was 3.5 percent in the hormone group and 4.2 percent in the placebo group (hazard ratio [HR] 0.79, 95% CI 0.7-0.9). There was little change in the HR after adjustment for changes in body mass index and waist circumference.

- Changes in fasting glucose and insulin during the first year of follow-up suggested a decrease in insulin resistance in the hormone group.

In a subsequent analysis, the estimated benefit for women ages 50 to 59 years, the group most likely to be taking MHT, was 11 fewer cases of type 2 diabetes per 1000 per five years of use (for estrogen-progestin and estrogen use combined) [6]. (See '[Women 50 to 59 years](#)' above.)

OTHER POSSIBLE BENEFITS

Recurrent urinary tract infection — There is evidence that vaginal ([figure 3](#)), but not systemic, estrogen therapy reduces the frequency of recurrent urinary tract infections in postmenopausal women [73-76]. (See "[Genitourinary syndrome of menopause: Treatment of vulvovaginal atrophy](#)", section on '[Vaginal estrogen therapy](#)' and "[Recurrent simple cystitis in female adults](#)".)

Health-related quality of life — In postmenopausal women with vasomotor flushes, estrogen appears to improve quality of life [77-80].

- In symptomatic women in the HERS trial, estrogen improved mental health and reduced depressive symptoms, whereas there were no changes in asymptomatic women [81,82].
- In symptomatic women in WHI, MHT relieved vasomotor flushes and improved sleep but did not otherwise improve quality of life [83].

Falls — Problems with balance may play a major role in the incidence of forearm fractures in postmenopausal women [84]. The incidence of Colles fractures increases markedly in women at age 50 years but remains stable in men up to the age of 80 years. A mechanism other than osteoporosis must be invoked to explain this observation because osteoporosis occurs gradually [85]. The fracture is usually caused by falling on an outstretched hand.

Estrogen therapy may improve balance and reduce the tendency to fall, a change that could contribute to the associated decrease in fracture risk. Some studies have noted an improvement in balance with estrogen therapy [86], while others have not [87,88]. In one study of 19 postmenopausal women with vasomotor symptoms, for example, dynamic posturography revealed that the women had improved balance during treatment with estrogen [86]. (See "[Menopausal hormone therapy in the prevention and treatment of osteoporosis](#)".)

Skin — Some clinicians believe that estrogen helps to preserve the thickness and the collagen content of skin in postmenopausal women [89-93], but two four-year, randomized trials have

reported no differences between MHT and placebo on age-related skin changes (wrinkling and skin rigidity) [94,95].

Although some epidemiologic studies have suggested that MHT is associated with an increased risk of melanoma [96,97], no excess risk of either melanoma or nonmelanoma skin cancer was observed in a post hoc analysis of the WHI (for unopposed estrogen or combined estrogen-progestin therapy) [98].

Eyes — Dry eye disease (DED) is thought to be related in part to estrogen deficiency; rates in women increase after menopause [99]. Administration of menopausal hormone therapy has had beneficial effects on the ocular surface in some [100], but not other studies [101]. However, in a meta-analysis of nine clinical trials in women with DED, improvements were seen in the symptoms of dryness, foreign body sensation, and burning compared with controls [102]. Unopposed estrogen was more effective than combined estrogen-progestin therapy, and women under age 55 years appeared to benefit more than those over age 55.

The risk of cataract formation may be decreased by long-term estrogen therapy. In the Framingham Heart Study, postmenopausal women who had taken estrogen for 10 years or longer had reduction in risk of nuclear lens opacities compared with control women (odds ratio [OR] 0.4, 95% CI 0.2-1.01) [103]. In a second study, (a survey) the prevalence of subcapsular as well as nuclear opacities were lower in women taking estrogen [104]. (See "[Cataract in adults](#)".)

MHT has also been associated with lower intraocular pressures [100,105,106] and lower rates of primary open-angle glaucoma [107], but the absolute reduction in risk is small. In a retrospective analysis of claims data from women over age 50 years, the calculated absolute risks of developing primary open-angle glaucoma (after adjusting for age, ethnicity, and ocular comorbidities) for women taking four years of unopposed estrogen, combined estrogen-progestin, or no HT were 1.6, 1.7, and 2.1 percent, respectively. We do not consider this small reduction in glaucoma risk to be an important consideration in deciding whether to recommend short-term HT for menopausal symptoms.

Other

- **Osteoarthritis** – It has been thought that estrogen may reduce the risk of osteoarthritis. In a cross-sectional study of over 4000 women, those receiving long-term estrogen therapy (≥ 10 years) had a 40 percent lower risk of hip osteoarthritis than those who had not received estrogen [108]. Similar results were found in a prospective analysis of the Framingham cohort [109]. In the WHI, women who took unopposed estrogen had lower rates of subsequent arthroplasty when compared with those taking placebo [110].
- **Dental** – Estrogen appears to preserve teeth [111,112]; the relative risk (RR) for edentia was only 0.6 in postmenopausal women taking estrogen as compared with those who did not,

perhaps related to less osteoporosis of the jaw [111].

OTHER POSSIBLE RISKS

Estrogen therapy may be associated with increases in the incidence of systemic lupus erythematosus (SLE), and the Raynaud phenomenon. The data are insufficient for epilepsy. Although data are conflicting, there have been concerns that estrogen therapy is associated with an increased risk of meningiomas. However, the reported risk is extremely low (2 cases per 10,000 over 5 years) [113].

- **Urinary incontinence** – Estrogen may alleviate dyspareunia, recurrent cystitis, and vaginal/urethral atrophy and inflammation in postmenopausal women. However, both the Heart and Estrogen/Progestin Replacement Study (HERS) and Women's Health Initiative (WHI) trials have reported that oral MHT worsens incontinence. Thus, oral estrogen with or without progestin should not be prescribed for this indication. Of note, extremely low doses of unopposed transdermal estrogen (0.014 mg/day) do not appear to increase the risk of urinary incontinence [114]. (See "[Female urinary incontinence: Treatment](#)", section on '[Topical vaginal estrogen](#)'.)

The use of topical vaginal estrogen therapy for urogenital atrophy symptoms is discussed separately. (See "[Genitourinary syndrome of menopause: Treatment of vulvovaginal atrophy](#)".)

- **Asthma** – Estrogen therapy may be associated with the onset of asthma. In the Nurses' Health Study, for example, the relative risk (RR) of new-onset asthma in 36,094 postmenopausal women followed for 10 years was significantly greater in women taking estrogen as compared with those who were not (RR 1.5) [115]. This increased risk was dose related; it was statistically significant only at a dose greater than 0.625 mg/day of [conjugated estrogens](#), probably because of the small numbers of women studied.

Data are conflicting on whether estrogen therapy in postmenopausal asthmatic women causes a worsening of airway function. In one study of 15 postmenopausal women with mild to moderate asthma, the estrogen-treated women had subclinical worsening of disease activity (as measured by peak expiratory flow and spirometry) [116]. In contrast, in a second study of 20 postmenopausal asthmatic women, there were no differences in measures of airway obstruction after stopping and restarting estrogen therapy [117].

Thus, while estrogen is not contraindicated in women with obstructive lung disease, clinicians should be aware of the possibility of worsening bronchospasm. Furthermore, estrogen may be considered as an etiologic factor in women who develop asthma during therapy.

- **Systemic lupus erythematosus** – Data to implicate estrogen in the pathogenesis of SLE are mixed. This topic is reviewed in detail elsewhere. (See ["Systemic lupus erythematosus: Epidemiology and pathogenesis"](#), section on 'Hormonal factors'.)
- **Uterine leiomyomas** – Use of MHT in the postreproductive years may cause some women with leiomyomas to continue to have symptoms after menopause. The risk of symptoms may depend, in part, on the location of the fibroid (higher if submucosal [118]) and type of estrogen preparation (higher with transdermal estrogen in some studies [119,120] but not others [121]). (See ["Uterine fibroids \(leiomyomas\): Epidemiology, clinical features, diagnosis, and natural history"](#), section on 'Reproductive and endocrine factors'.)

A systematic review including five randomized controlled trials found that MHT caused myoma growth, but this typically occurred without clinical symptoms [122]. These findings were confirmed in a subsequent prospective study [123]. Thus, presence of leiomyomas is not a contraindication to MHT nor associated with new symptomatic fibroids in most women.

OTHER ISSUES

Weight — Although women are often concerned that taking MHT will exacerbate the weight gain that occurs in midlife, a meta-analysis of 28 trials in 28,559 women found no evidence of an effect of unopposed estrogen or combined estrogen-progestin on body weight or body mass index [124].

Early menopause and primary ovarian insufficiency — Data from the Women's Health Initiative (WHI) should not be extrapolated to women with early menopause or primary ovarian insufficiency (menopause between ages 40 and 45 years or before age 40 years, respectively) in whom MHT should be initiated at a younger age. In otherwise healthy women with primary ovarian insufficiency, we continue their MHT until the average age of menopause, approximately age 50 to 51 years. At that point, the same discussion of potential risks and benefits of MHT should take place. (See ["Management of primary ovarian insufficiency \(premature ovarian failure\)"](#).)

Androgen therapy — The use of exogenous androgen therapy in peri- and postmenopausal women is reviewed separately. (See ["Overview of sexual dysfunction in females: Management"](#), section on 'Androgens'.)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See ["Society guideline links: Menopause"](#).)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see "[Patient education: Hormone therapy for menopause \(The Basics\)](#)" and "[Patient education: Menopause \(The Basics\)](#)")
- Beyond the Basics topics (see "[Patient education: Menopause \(Beyond the Basics\)](#)" and "[Patient education: Menopausal hormone therapy \(Beyond the Basics\)](#)" and "[Patient education: Non-estrogen treatments for menopausal symptoms \(Beyond the Basics\)](#)" and "[Patient education: Vaginal dryness \(Beyond the Basics\)](#)")

SUMMARY AND RECOMMENDATIONS

- **General principles** – During the menopause transition and postmenopause, most women experience vasomotor symptoms (hot flashes and night sweats). Systemic estrogen is the most effective treatment available for relief of hot flashes. Other symptoms associated with perimenopause and menopause that respond to systemic estrogen include sleep disturbances, mood lability/depression, and, in some cases, joint aches and pains. Estrogen is no longer recommended for prevention of chronic disease such as coronary heart disease (CHD) or osteoporosis, preservation of cognitive function, or prevention of dementia. (See '[Introduction](#)' above.)
- **Estimates of overall risk based on age**
 - **Individualized risk assessment** – When counseling symptomatic women considering MHT, the clinician should provide an individualized assessment of the potential risks and benefits for up to five years of use based upon age-specific data. (See '[Individualized risk assessment](#)' above.)

- **Women ≥ 60 years of age** – It is particularly important to individualize the assessment in women 60 to 69 years of age. The Women's Health Initiative (WHI) and subsequent analyses show an excess risk of vascular events with [conjugated equine estrogen](#) (with or without [medroxyprogesterone acetate](#)) compared with placebo in older postmenopausal women (≥ 70 years). The risk in women 60 to 69 years of age is lower, but risk assessment should be individualized. (See '[Women \$\geq 60\$ years](#)' above and '[Cardiovascular effects](#)' above.)
 - **Women 50 to 59 years of age** – The overall risks of MHT in postmenopausal women 50 to 59 years of age are considerably lower than those for older women. The major explanation for the difference in absolute excess risk between older and younger postmenopausal women is the lower baseline risk of CHD, stroke, venous thromboembolism (VTE), and breast cancer in younger postmenopausal women. Women should be reassured that the absolute risk of complications for healthy, young postmenopausal women taking MHT is very low ([figure 1](#)). (See '[Women 50 to 59 years](#)' above and '[Cardiovascular effects](#)' above.)
 - **Mortality** – Available data suggest that MHT is not associated with an increase in mortality rates when compared with placebo. Younger menopausal women (ages 50 to 59) may actually have a reduction in all-cause mortality with MHT. (See '[Mortality](#)' above.)
 - **Breast cancer** – In the WHI overall study population, combined [conjugated equine estrogen](#) (CEE)-medroxyprogesterone acetate use compared with placebo increased the risk of breast cancer. However, the absolute risk of breast cancer with combined menopausal hormone therapy is very low with typical duration of use in postmenopausal women ages 50 to 59 years (eg, three additional cases per 1000 women with CEE plus [medroxyprogesterone acetate](#) for five years, therefore <1 per 1000/year) ([figure 1](#)).
- In the WHI overall study population, unopposed estrogen (CEE) reduced the risk of breast cancer. For women in their 50s taking unopposed CEE, there were an estimated 2.5 fewer cases per 1000 women per five years of hormone use when compared with placebo ([figure 1](#)).
- **Endometrial cancer** – In patients with a uterus, systemic estrogen therapy from any route (oral, transdermal patch, vaginal ring) without concurrent administration of a progestogen increases the risk of developing endometrial hyperplasia and carcinoma. Therefore, all women with an intact uterus need to take a progestogen with systemic estrogen. (See '[Endometrial carcinoma: Epidemiology, risk factors, and prevention](#)', section on '[Chronically increased estrogen levels or estrogenic activity](#)'.)

REFERENCES

1. "The 2022 Hormone Therapy Position Statement of The North American Menopause Society" Advisory Panel. The 2022 hormone therapy position statement of The North American Menopause Society. *Menopause* 2022; 29:767.
2. Stuenkel CA, Davis SR, Gompel A, et al. Treatment of Symptoms of the Menopause: An Endocrine Society Clinical Practice Guideline. *J Clin Endocrinol Metab* 2015; 100:3975.
3. US Preventive Services Task Force, Mangione CM, Barry MJ, et al. Hormone Therapy for the Primary Prevention of Chronic Conditions in Postmenopausal Persons: US Preventive Services Task Force Recommendation Statement. *JAMA* 2022; 328:1740.
4. By the 2023 American Geriatrics Society Beers Criteria® Update Expert Panel. American Geriatrics Society 2023 updated AGS Beers Criteria® for potentially inappropriate medication use in older adults. *J Am Geriatr Soc* 2023; 71:2052.
5. Manson JE, Chlebowski RT, Stefanick ML, et al. Menopausal hormone therapy and health outcomes during the intervention and extended poststopping phases of the Women's Health Initiative randomized trials. *JAMA* 2013; 310:1353.
6. Santen RJ, Allred DC, Ardoin SP, et al. Postmenopausal hormone therapy: an Endocrine Society scientific statement. *J Clin Endocrinol Metab* 2010; 95:s1.
7. Rossouw JE, Aragaki AK, Manson JE, et al. Menopausal Hormone Therapy and Cardiovascular Diseases in Women With Vasomotor Symptoms: A Secondary Analysis of the Women's Health Initiative Randomized Clinical Trials. *JAMA Intern Med* 2025; 185:1330.
8. Rossouw JE, Anderson GL, Prentice RL, et al. Risks and benefits of estrogen plus progestin in healthy postmenopausal women: principal results From the Women's Health Initiative randomized controlled trial. *JAMA* 2002; 288:321.
9. Anderson GL, Limacher M, Assaf AR, et al. Effects of conjugated equine estrogen in postmenopausal women with hysterectomy: the Women's Health Initiative randomized controlled trial. *JAMA* 2004; 291:1701.
10. Marjoribanks J, Farquhar C, Roberts H, et al. Long-term hormone therapy for perimenopausal and postmenopausal women. *Cochrane Database Syst Rev* 2017; 1:CD004143.
11. Gartlehner G, Patel SV, Reddy S, et al. Hormone Therapy for the Primary Prevention of Chronic Conditions in Postmenopausal Persons: Updated Evidence Report and Systematic Review for the US Preventive Services Task Force. *JAMA* 2022; 328:1747.
12. ACOG Practice Bulletin No. 141: management of menopausal symptoms. *Obstet Gynecol* 2014; 123:202. Reaffirmed 2018.
13. Hulley S, Grady D, Bush T, et al. Randomized trial of estrogen plus progestin for secondary prevention of coronary heart disease in postmenopausal women. *Heart and*

- Estrogen/progestin Replacement Study (HERS) Research Group. JAMA 1998; 280:605.
14. Grady D, Herrington D, Bittner V, et al. Cardiovascular disease outcomes during 6.8 years of hormone therapy: Heart and Estrogen/progestin Replacement Study follow-up (HERS II). JAMA 2002; 288:49.
 15. Waters DD, Alderman EL, Hsia J, et al. Effects of hormone replacement therapy and antioxidant vitamin supplements on coronary atherosclerosis in postmenopausal women: a randomized controlled trial. JAMA 2002; 288:2432.
 16. Rossouw JE, Prentice RL, Manson JE, et al. Postmenopausal hormone therapy and risk of cardiovascular disease by age and years since menopause. JAMA 2007; 297:1465.
 17. Manson JE, Hsia J, Johnson KC, et al. Estrogen plus progestin and the risk of coronary heart disease. N Engl J Med 2003; 349:523.
 18. Shufelt CL, Johnson BD, Berga SL, et al. Timing of hormone therapy, type of menopause, and coronary disease in women: data from the National Heart, Lung, and Blood Institute-sponsored Women's Ischemia Syndrome Evaluation. Menopause 2011; 18:943.
 19. Manson JE, Allison MA, Rossouw JE, et al. Estrogen therapy and coronary-artery calcification. N Engl J Med 2007; 356:2591.
 20. Salpeter SR, Walsh JM, Greyber E, Salpeter EE. Brief report: Coronary heart disease events associated with hormone therapy in younger and older women. A meta-analysis. J Gen Intern Med 2006; 21:363.
 21. Harman SM, Black DM, Naftolin F, et al. Arterial imaging outcomes and cardiovascular risk factors in recently menopausal women: a randomized trial. Ann Intern Med 2014; 161:249.
 22. Hodis HN, Mack WJ, Henderson VW, et al. Vascular Effects of Early versus Late Postmenopausal Treatment with Estradiol. N Engl J Med 2016; 374:1221.
 23. Boardman HM, Hartley L, Eisinga A, et al. Hormone therapy for preventing cardiovascular disease in post-menopausal women. Cochrane Database Syst Rev 2015; :CD002229.
 24. Schierbeck LL, Rejnmark L, Tofteng CL, et al. Effect of hormone replacement therapy on cardiovascular events in recently postmenopausal women: randomised trial. BMJ 2012; 345:e6409.
 25. Wassertheil-Smoller S, Hendrix SL, Limacher M, et al. Effect of estrogen plus progestin on stroke in postmenopausal women: the Women's Health Initiative: a randomized trial. JAMA 2003; 289:2673.
 26. Simon JA, Hsia J, Cauley JA, et al. Postmenopausal hormone therapy and risk of stroke: The Heart and Estrogen-progestin Replacement Study (HERS). Circulation 2001; 103:638.
 27. Viscoli CM, Brass LM, Kernan WN, et al. A clinical trial of estrogen-replacement therapy after ischemic stroke. N Engl J Med 2001; 345:1243.

28. Renoux C, Dell'aniello S, Garbe E, Suissa S. Transdermal and oral hormone replacement therapy and the risk of stroke: a nested case-control study. *BMJ* 2010; 340:c2519.
29. Gu Y, Han F, Xue M, et al. The benefits and risks of menopause hormone therapy for the cardiovascular system in postmenopausal women: a systematic review and meta-analysis. *BMC Womens Health* 2024; 24:60.
30. Cushman M, Kuller LH, Prentice R, et al. Estrogen plus progestin and risk of venous thrombosis. *JAMA* 2004; 292:1573.
31. Canonico M, Plu-Bureau G, Lowe GD, Scarabin PY. Hormone replacement therapy and risk of venous thromboembolism in postmenopausal women: systematic review and meta-analysis. *BMJ* 2008; 336:1227.
32. Hicks A, Robson D, Tellis B, et al. Safety of menopause hormone therapy in postmenopausal women at higher risk of venous thromboembolism: a systematic review. *Climacteric* 2025; 28:497.
33. Renoux C, Dell'Aniello S, Suissa S. Hormone replacement therapy and the risk of venous thromboembolism: a population-based study. *J Thromb Haemost* 2010; 8:979.
34. Sweetland S, Beral V, Balkwill A, et al. Venous thromboembolism risk in relation to use of different types of postmenopausal hormone therapy in a large prospective study. *J Thromb Haemost* 2012; 10:2277.
35. Walsh BW, Schiff I, Rosner B, et al. Effects of postmenopausal estrogen replacement on the concentrations and metabolism of plasma lipoproteins. *N Engl J Med* 1991; 325:1196.
36. Darling GM, Johns JA, McCloud PI, Davis SR. Estrogen and progestin compared with simvastatin for hypercholesterolemia in postmenopausal women. *N Engl J Med* 1997; 337:595.
37. Binder EF, Williams DB, Schechtman KB, et al. Effects of hormone replacement therapy on serum lipids in elderly women. a randomized, placebo-controlled trial. *Ann Intern Med* 2001; 134:754.
38. Ory SJ, Field CS, Herrmann RR, et al. Effects of long-term transdermal administration of estradiol on serum lipids. *Mayo Clin Proc* 1998; 73:735.
39. Manson JE, Aragaki AK, Rossouw JE, et al. Menopausal Hormone Therapy and Long-term All-Cause and Cause-Specific Mortality: The Women's Health Initiative Randomized Trials. *JAMA* 2017; 318:927.
40. Hulley S, Furberg C, Barrett-Connor E, et al. Noncardiovascular disease outcomes during 6.8 years of hormone therapy: Heart and Estrogen/progestin Replacement Study follow-up (HERS II). *JAMA* 2002; 288:58.
41. Salpeter SR, Cheng J, Thabane L, et al. Bayesian meta-analysis of hormone therapy and mortality in younger postmenopausal women. *Am J Med* 2009; 122:1016.

42. Martin KA, Manson JE. Approach to the patient with menopausal symptoms. *J Clin Endocrinol Metab* 2008; 93:4567.
43. Chlebowski RT, Aragaki AK, Pan K, et al. Randomized trials of estrogen-alone and breast cancer incidence: a meta-analysis. *Breast Cancer Res Treat* 2024; 206:177.
44. Collaborative Group on Hormonal Factors in Breast Cancer. Type and timing of menopausal hormone therapy and breast cancer risk: individual participant meta-analysis of the worldwide epidemiological evidence. *Lancet* 2019; 394:1159.
45. Chen WY, Manson JE, Hankinson SE, et al. Unopposed estrogen therapy and the risk of invasive breast cancer. *Arch Intern Med* 2006; 166:1027.
46. The NAMS 2017 Hormone Therapy Position Statement Advisory Panel. The 2017 hormone therapy position statement of The North American Menopause Society. *Menopause* 2017; 24:728.
47. Abenhaim HA, Suissa S, Azoulay L, et al. Menopausal Hormone Therapy Formulation and Breast Cancer Risk. *Obstet Gynecol* 2022; 139:1103.
48. Fournier A, Berrino F, Clavel-Chapelon F. Unequal risks for breast cancer associated with different hormone replacement therapies: results from the E3N cohort study. *Breast Cancer Res Treat* 2008; 107:103.
49. Anderson GL, Judd HL, Kaunitz AM, et al. Effects of estrogen plus progestin on gynecologic cancers and associated diagnostic procedures: the Women's Health Initiative randomized trial. *JAMA* 2003; 290:1739.
50. Collaborative Group On Epidemiological Studies Of Ovarian Cancer, Beral V, Gaitskell K, et al. Menopausal hormone use and ovarian cancer risk: individual participant meta-analysis of 52 epidemiological studies. *Lancet* 2015; 385:1835.
51. Kotsopoulos J, Lubinski J, Neuhausen SL, et al. Hormone replacement therapy and the risk of ovarian cancer in BRCA1 and BRCA2 mutation carriers. *Gynecol Oncol* 2006; 100:83.
52. Johnson SR, Ettinger B, Macer JL, et al. Uterine and vaginal effects of unopposed ultralow-dose transdermal estradiol. *Obstet Gynecol* 2005; 105:779.
53. Woodruff JD, Pickar JH. Incidence of endometrial hyperplasia in postmenopausal women taking conjugated estrogens (Premarin) with medroxyprogesterone acetate or conjugated estrogens alone. The Menopause Study Group. *Am J Obstet Gynecol* 1994; 170:1213.
54. Beresford SA, Weiss NS, Voigt LF, McKnight B. Risk of endometrial cancer in relation to use of oestrogen combined with cyclic progestagen therapy in postmenopausal women. *Lancet* 1997; 349:458.
55. Effects of hormone replacement therapy on endometrial histology in postmenopausal women. The Postmenopausal Estrogen/Progestin Interventions (PEPI) Trial. The Writing

Group for the PEPI Trial. JAMA 1996; 275:370.

56. Lacey JV Jr, Leitzmann MF, Chang SC, et al. Endometrial cancer and menopausal hormone therapy in the National Institutes of Health-AARP Diet and Health Study cohort. Cancer 2007; 109:1303.
57. Furness S, Roberts H, Marjoribanks J, Lethaby A. Hormone therapy in postmenopausal women and risk of endometrial hyperplasia. Cochrane Database Syst Rev 2012; :CD000402.
58. Weiderpass E, Adami HO, Baron JA, et al. Risk of endometrial cancer following estrogen replacement with and without progestins. J Natl Cancer Inst 1999; 91:1131.
59. Pike MC, Peters RK, Cozen W, et al. Estrogen-progestin replacement therapy and endometrial cancer. J Natl Cancer Inst 1997; 89:1110.
60. Cerin A, Heldaas K, Moeller B. Adverse endometrial effects of long-cycle estrogen and progestogen replacement therapy. The Scandinavian LongCycle Study Group. N Engl J Med 1996; 334:668.
61. Whitehead MI, Fraser D. The effects of estrogens and progestogens on the endometrium. Modern approach to treatment. Obstet Gynecol Clin North Am 1987; 14:299.
62. Whitehead MI, Hillard TC, Crook D. The role and use of progestogens. Obstet Gynecol 1990; 75:59S.
63. Chlebowski RT, Wactawski-Wende J, Ritenbaugh C, et al. Estrogen plus progestin and colorectal cancer in postmenopausal women. N Engl J Med 2004; 350:991.
64. Chlebowski RT, Aragaki AK, Pan K, et al. Estrogen Plus Progestin and Colorectal Cancer: Long-Term Findings From the Women's Health Initiative Randomized Clinical Trial. J Clin Oncol 2024; 42:3530.
65. O'Brien J, Jackson JW, Grodstein F, et al. Postmenopausal hormone therapy is not associated with risk of all-cause dementia and Alzheimer's disease. Epidemiol Rev 2014; 36:83.
66. Lethaby A, Hogervorst E, Richards M, et al. Hormone replacement therapy for cognitive function in postmenopausal women. Cochrane Database Syst Rev 2008; :CD003122.
67. Melville M, He L, Desai R, et al. Menopause hormone therapy and risk of mild cognitive impairment or dementia: a systematic review and meta-analysis. Lancet Healthy Longev 2025; 6:100803.
68. Andy C, Nerattini M, Jett S, et al. Systematic review and meta-analysis of the effects of menopause hormone therapy on cognition. Front Endocrinol (Lausanne) 2024; 15:1350318.
69. Cirillo DJ, Wallace RB, Rodabough RJ, et al. Effect of estrogen therapy on gallbladder disease. JAMA 2005; 293:330.

70. Simon JA, Hunninghake DB, Agarwal SK, et al. Effect of estrogen plus progestin on risk for biliary tract surgery in postmenopausal women with coronary artery disease. The Heart and Estrogen/progestin Replacement Study. *Ann Intern Med* 2001; 135:493.
71. Kanaya AM, Herrington D, Vittinghoff E, et al. Glycemic effects of postmenopausal hormone therapy: the Heart and Estrogen/progestin Replacement Study. A randomized, double-blind, placebo-controlled trial. *Ann Intern Med* 2003; 138:1.
72. Margolis KL, Bonds DE, Rodabough RJ, et al. Effect of oestrogen plus progestin on the incidence of diabetes in postmenopausal women: results from the Women's Health Initiative Hormone Trial. *Diabetologia* 2004; 47:1175.
73. Raz R, Stamm WE. A controlled trial of intravaginal estriol in postmenopausal women with recurrent urinary tract infections. *N Engl J Med* 1993; 329:753.
74. Cardozo L, Lose G, McClish D, et al. A systematic review of estrogens for recurrent urinary tract infections: third report of the hormones and urogenital therapy (HUT) committee. *Int Urogynecol J Pelvic Floor Dysfunct* 2001; 12:15.
75. Brown JS, Vittinghoff E, Kanaya AM, et al. Urinary tract infections in postmenopausal women: effect of hormone therapy and risk factors. *Obstet Gynecol* 2001; 98:1045.
76. Stamm WE. Estrogens and urinary-tract infection. *J Infect Dis* 2007; 195:623.
77. MacLennan A, Lester S, Moore V. Oral estrogen replacement therapy versus placebo for hot flushes: a systematic review. *Climacteric* 2001; 4:58.
78. Col NF, Weber G, Stiggelbout A, et al. Short-term menopausal hormone therapy for symptom relief: an updated decision model. *Arch Intern Med* 2004; 164:1634.
79. Tang R, Fan Y, Ruan X, et al. Changes in menopause-specific quality of life between women with transdermal estradiol versus oral estrogens: results of a randomized controlled trial. *Gynecol Endocrinol* 2025; 41:2484213.
80. Utian WH, Woods NF. Impact of hormone therapy on quality of life after menopause. *Menopause* 2013; 20:1098.
81. Hlatky MA, Boothroyd D, Vittinghoff E, et al. Quality-of-life and depressive symptoms in postmenopausal women after receiving hormone therapy: results from the Heart and Estrogen/Progestin Replacement Study (HERS) trial. *JAMA* 2002; 287:591.
82. Rexrode KM, Manson JE. Postmenopausal hormone therapy and quality of life: no cause for celebration. *JAMA* 2002; 287:641.
83. Brunner RL, Gass M, Aragaki A, et al. Effects of conjugated equine estrogen on health-related quality of life in postmenopausal women with hysterectomy: results from the Women's Health Initiative Randomized Clinical Trial. *Arch Intern Med* 2005; 165:1976.
84. Ekblad S, Bergendahl A, Enler P, et al. Disturbances in postural balance are common in postmenopausal women with vasomotor symptoms. *Climacteric* 2000; 3:192.

85. Crilly RG, Delaquerrière Richardson L, Roth JH, et al. Postural stability and Colles' fracture. *Age Ageing* 1987; 16:133.
86. Hammar ML, Lindgren R, Berg GE, et al. Effects of hormonal replacement therapy on the postural balance among postmenopausal women. *Obstet Gynecol* 1996; 88:955.
87. Seeley DG, Cauley JA, Grady D, et al. Is postmenopausal estrogen therapy associated with neuromuscular function or falling in elderly women? Study of Osteoporotic Fractures Research Group. *Arch Intern Med* 1995; 155:293.
88. Armstrong AL, Osborne J, Coupland CA, et al. Effects of hormone replacement therapy on muscle performance and balance in post-menopausal women. *Clin Sci (Lond)* 1996; 91:685.
89. Brincat M, Versi E, Moniz CF, et al. Skin collagen changes in postmenopausal women receiving different regimens of estrogen therapy. *Obstet Gynecol* 1987; 70:123.
90. Holland EF, Studd JW, Mansell JP, et al. Changes in collagen composition and cross-links in bone and skin of osteoporotic postmenopausal women treated with percutaneous estradiol implants. *Obstet Gynecol* 1994; 83:180.
91. Maheux R, Naud F, Rioux M, et al. A randomized, double-blind, placebo-controlled study on the effect of conjugated estrogens on skin thickness. *Am J Obstet Gynecol* 1994; 170:642.
92. Ashcroft GS, Dodsworth J, van Boxtel E, et al. Estrogen accelerates cutaneous wound healing associated with an increase in TGF-beta1 levels. *Nat Med* 1997; 3:1209.
93. Wolff EF, Narayan D, Taylor HS. Long-term effects of hormone therapy on skin rigidity and wrinkles. *Fertil Steril* 2005; 84:285.
94. Phillips TJ, Symons J, Menon S, HT Study Group. Does hormone therapy improve age-related skin changes in postmenopausal women? A randomized, double-blind, double-dummy, placebo-controlled multicenter study assessing the effects of norethindrone acetate and ethinyl estradiol in the improvement of mild to moderate age-related skin changes in postmenopausal women. *J Am Acad Dermatol* 2008; 59:397.
95. Owen CM, Pal L, Mumford SL, et al. Effects of hormones on skin wrinkles and rigidity vary by race/ethnicity: four-year follow-up from the ancillary skin study of the Kronos Early Estrogen Prevention Study. *Fertil Steril* 2016; 106:1170.
96. Koomen ER, Joosse A, Herings RM, et al. Estrogens, oral contraceptives and hormonal replacement therapy increase the incidence of cutaneous melanoma: a population-based case-control study. *Ann Oncol* 2009; 20:358.
97. Holly EA, Cress RD, Ahn DK. Cutaneous melanoma in women: ovulatory life, menopause, and use of exogenous estrogens. *Cancer Epidemiol Biomarkers Prev* 1994; 3:661.
98. Tang JY, Spaunhurst KM, Chlebowski RT, et al. Menopausal hormone therapy and risks of melanoma and nonmelanoma skin cancers: women's health initiative randomized trials. *J*

Natl Cancer Inst 2011; 103:1469.

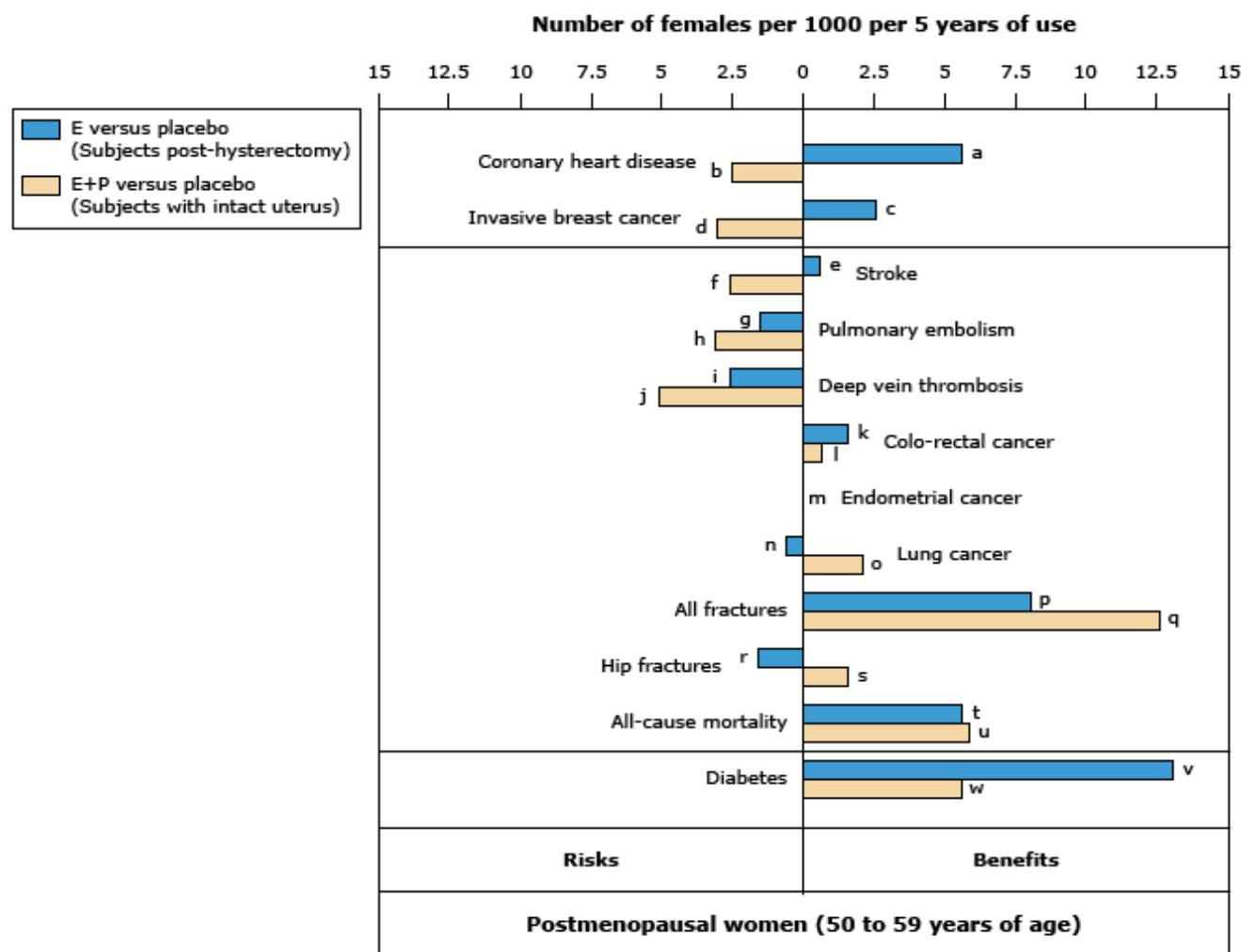
99. Sullivan DA, Rocha EM, Aragona P, et al. TFOS DEWS II Sex, Gender, and Hormones Report. *Ocul Surf* 2017; 15:284.
100. Affinito P, Di Spiezio Sardo A, Di Carlo C, et al. Effects of hormone replacement therapy on ocular function in postmenopause. *Menopause* 2003; 10:482.
101. Schaumberg DA, Buring JE, Sullivan DA, Dana MR. Hormone replacement therapy and dry eye syndrome. *JAMA* 2001; 286:2114.
102. Hao Y, Xiaodan J, Jiarui Y, Xuemin L. The effect of hormone therapy on the ocular surface and intraocular pressure for postmenopausal women: a systematic review and meta-analysis of randomized controlled trials. *Menopause* 2020; 27:929.
103. Worzala K, Hiller R, Sperduto RD, et al. Postmenopausal estrogen use, type of menopause, and lens opacities: the Framingham studies. *Arch Intern Med* 2001; 161:1448.
104. Freeman EE, Munoz B, Schein OD, West SK. Hormone replacement therapy and lens opacities: the Salisbury Eye Evaluation project. *Arch Ophthalmol* 2001; 119:1687.
105. Tint NL, Alexander P, Tint KM, et al. Hormone therapy and intraocular pressure in nonglaucomatous eyes. *Menopause* 2010; 17:157.
106. Vajaranant TS, Maki PM, Pasquale LR, et al. Effects of Hormone Therapy on Intraocular Pressure: The Women's Health Initiative-Sight Exam Study. *Am J Ophthalmol* 2016; 165:115.
107. Newman-Casey PA, Talwar N, Nan B, et al. The potential association between postmenopausal hormone use and primary open-angle glaucoma. *JAMA Ophthalmol* 2014; 132:298.
108. Nevitt MC, Cummings SR, Lane NE, et al. Association of estrogen replacement therapy with the risk of osteoarthritis of the hip in elderly white women. Study of Osteoporotic Fractures Research Group. *Arch Intern Med* 1996; 156:2073.
109. Zhang Y, McAlindon TE, Hannan MT, et al. Estrogen replacement therapy and worsening of radiographic knee osteoarthritis: the Framingham Study. *Arthritis Rheum* 1998; 41:1867.
110. Cirillo DJ, Wallace RB, Wu L, Yood RA. Effect of hormone therapy on risk of hip and knee joint replacement in the Women's Health Initiative. *Arthritis Rheum* 2006; 54:3194.
111. Paganini-Hill A. The benefits of estrogen replacement therapy on oral health. The Leisure World cohort. *Arch Intern Med* 1995; 155:2325.
112. Krall EA, Dawson-Hughes B, Hannan MT, et al. Postmenopausal estrogen replacement and tooth retention. *Am J Med* 1997; 102:536.
113. Benson VS, Kirichek O, Beral V, Green J. Menopausal hormone therapy and central nervous system tumor risk: large UK prospective study and meta-analysis. *Int J Cancer*

2015; 136:2369.

114. Waetjen LE, Brown JS, Vittinghoff E, et al. The effect of ultralow-dose transdermal estradiol on urinary incontinence in postmenopausal women. *Obstet Gynecol* 2005; 106:946.
115. Troisi RJ, Speizer FE, Willett WC, et al. Menopause, postmenopausal estrogen preparations, and the risk of adult-onset asthma. A prospective cohort study. *Am J Respir Crit Care Med* 1995; 152:1183.
116. Lieberman D, Kopernik G, Porath A, et al. Sub-clinical worsening of bronchial asthma during estrogen replacement therapy in asthmatic post-menopausal women. *Maturitas* 1995; 21:153.
117. Hepburn MJ, Dooley DP, Morris MJ. The effects of estrogen replacement therapy on airway function in postmenopausal, asthmatic women. *Arch Intern Med* 2001; 161:2717.
118. Akkad AA, Habiba MA, Ismail N, et al. Abnormal uterine bleeding on hormone replacement: the importance of intrauterine structural abnormalities. *Obstet Gynecol* 1995; 86:330.
119. Sener AB, Seçkin NC, Ozmen S, et al. The effects of hormone replacement therapy on uterine fibroids in postmenopausal women. *Fertil Steril* 1996; 65:354.
120. Polatti F, Viazzo F, Colleoni R, Nappi RE. Uterine myoma in postmenopause: a comparison between two therapeutic schedules of HRT. *Maturitas* 2000; 37:27.
121. Palomba S, Sena T, Noia R, et al. Transdermal hormone replacement therapy in postmenopausal women with uterine leiomyomas. *Obstet Gynecol* 2001; 98:1053.
122. Ang WC, Farrell E, Vollenhoven B. Effect of hormone replacement therapies and selective estrogen receptor modulators in postmenopausal women with uterine leiomyomas: a literature review. *Climacteric* 2001; 4:284.
123. Yang CH, Lee JN, Hsu SC, et al. Effect of hormone replacement therapy on uterine fibroids in postmenopausal women--a 3-year study. *Maturitas* 2002; 43:35.
124. Norman RJ, Flight IH, Rees MC. Oestrogen and progestogen hormone replacement therapy for peri-menopausal and post-menopausal women: weight and body fat distribution. *Cochrane Database Syst Rev* 2000; :CD001018.

GRAPHICS

Figure 1: Risks and benefits of menopausal hormone therapy (MHT)



Updated summary of the effects of orally administered CEE alone (subjects post-hysterectomy) or combined with MPA (subjects with an intact uterus) in females ages 50 to 59 years during intervention phase of WHI. One set of analyses examined the risks and benefits of these agents in females ages 50 to 59 years. This figure plots these data, which are expressed here as excess risks and benefits per 1000 females using MHT for 5 years. Because females deciding to use MHT are more likely to continue this for a period of years rather than 1 year, this figure is constructed according to that assumption. WHI studies were not powered for age-related subset analyses, and none of the data presented in the figure are statistically significant. Nonetheless, this figure represents the best estimates that are available at the present time and are likely more reliable than similar estimates based on observational studies as reported previously in The Endocrine Society Scientific Statement.

For endometrial cancer risk, no colored bars are shown because no increase or decrease in risk was seen in either treatment group compared with placebo. All subjects in the CEE-alone group had undergone hysterectomy and therefore had no risk. Combined CEE-MPA therapy in subjects with an intact uterus was not associated with an increased risk, suggesting that the MPA regimen provided adequate protection against estrogen-associated endometrial cancer.

The HR (95% CI) values for the bars in the figure are listed here with reference to the alphabetical designations shown next to the bars: (a) HR 0.60 (0.35-1.04); (b) HR 1.34 (0.82-2.19); (c) HR 0.82 (0.50-1.34); (d) HR 1.21 (0.81-1.80); (e) HR 0.99 (0.53-1.85); (f) HR 1.51 (0.81-2.82); (g) HR 1.53 (0.63-3.75); (h)

HR 2.05 (0.89-4.71); (i) HR 1.66 (0.76-3.67); (j) HR 3.01 (1.36-6.66); (k) HR 0.71 (0.30-1.67); (l) HR 0.79 (0.29-2.18); (m) HR 1.00 (ns-ns); (n) HR 1.12 (0.45-2.75); (o) HR 0.62 (0.30-1.29); (p) HR 0.90 (0.72-1.11); (q) HR 0.82 (0.68-1.00); (r) HR 5.01 (0.59-42.9); (s) HR 0.17 (0.02-1.45); (t) HR 0.70 (0.46-1.09); (u) HR 0.67 (0.43-1.04); (v) HR 0.83 (0.67-1.04); and (w) HR 0.85 (0.66-1.09).^[1]

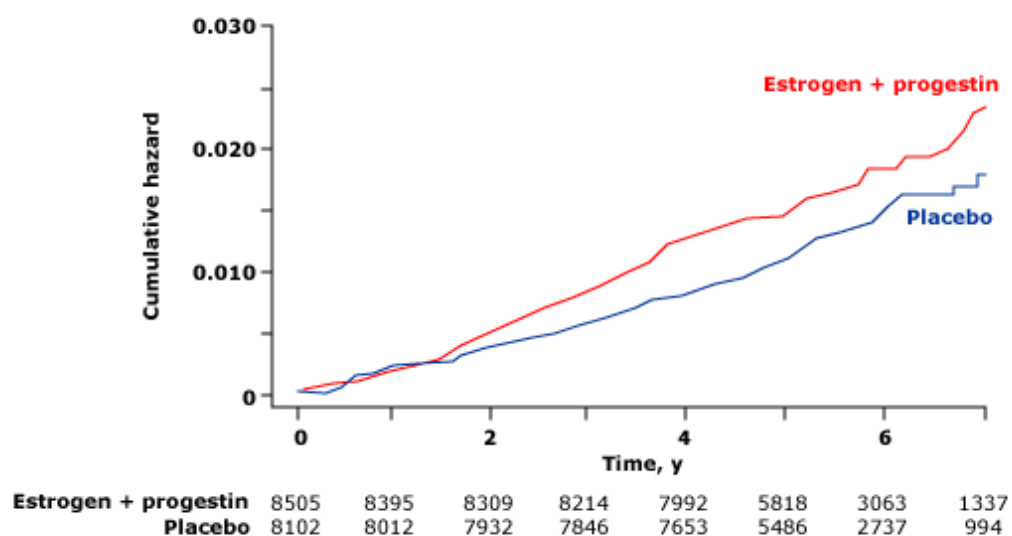
CEE: conjugated equine estrogen; E: estrogen; E+P: estrogen-progestin; HR: hazard ratio; MHT: menopausal hormone therapy; MPA: medroxyprogesterone acetate; WHI: Women's Health Initiative.

Reference:

1. Santen RJ, Stuenkel CA, Burger HG, Manson JE. Competency in menopause management: whither goest the internist? *J Womens Health (Larchmt)* 2014; 23:281.

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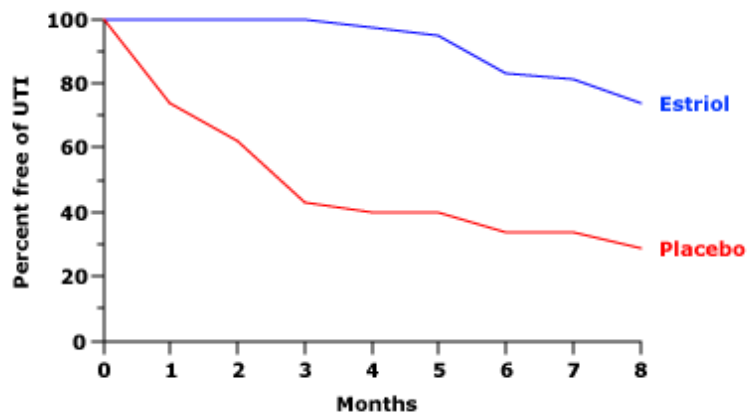
Figure 2: Women's Health Initiative: Risk of stroke



Kaplan-Meier estimates of cumulative hazard rates of stroke. In the Women's Health Initiative, combined estrogen-progestin therapy was associated with a significant increase in stroke when compared with placebo. The intention-to-treat hazard ratio was 1.31, 95% CI 1.02-1.68.

Data from: Wassertheil-Smoller S, Hendrix SL, Limacher M, et al. Effect of estrogen plus progestin on stroke in postmenopausal women: The Women's Health Initiative: A randomized trial. JAMA 2003; 289:2673.

Figure 3: Estrogen minimizes recurrent urinary tract infection



The effect of intravaginal estriol or placebo on the incidence of UTI in postmenopausal women with recurrent UTIs. Estrogen therapy was associated with a much greater likelihood of remaining free of infection.

UTI: urinary tract infection.

Data from: Raz R, Stamm WE. A controlled trial of intravaginal estriol in postmenopausal women with recurrent urinary tract infections. *N Engl J Med* 1993; 329:753.

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